Bryan Ramirez-Gonzalez

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EDUCATION

The University of Southern California — **Los Angeles, CA**
Bachelor of Science in Computer Science, Merit Scholar, Honors Engineering Track — *Expected May 2028*
- Relevant Coursework: Data Structures, Algorithms, Object-Oriented Programming, Embedded Systems, Discrete Mathematics, Linear Algebra

SKILLS

Programming Languages: Python, C, C++, Java, JavaScript/TypeScript, SQL, Swift, FastAPI, Flask, Node.js, React
AI/ML: PyTorch, LangGraph, LangChain, MCP, RAG (pgvector, BM25), Agent Evals, Computer Vision, NumPy, Pandas
Infrastructure: Kubernetes/OpenShift, Docker, Terraform, GCP, AWS, Temporal, PostgreSQL, GitHub Actions, Linux, pytest, Locust

EXPERIENCE

Google — Incoming *Software Engineering Intern* (Fall 2027)

NVIDIA — **Santa Clara, CA**
Software Engineering Intern (Summer 2026 & Fall 2026) — May 2026 - Present
- Designed and implemented a hybrid deterministic–LLM pre-flight gate for an ISO 26262 FMEDA safety-report tool (slated for 5–10 yrs of production use), closing a class of silent failures via 57 validation checks, AST code analysis, and human-in-the-loop review.
- Designed and built an 8-node LangGraph safety-review agent, leading cross-model regression testing on a 76-case known-defect eval set across 12 models (912 runs per campaign) and diagnosing model–prompt coupling to reach 100% recall (1,224/1,224 findings).
- Led verification of Codex and Claude Code agent plugins across 1,044 agent eval runs and 29 model/reasoning configurations.
- Deployed a containerized safety-review backend to Kubernetes/OpenShift for production migration, validated by 19 assertions.

Hudson River Trading — **New York, NY**
Undergraduate Fellow (Inside HRT 2026 — Quant Research Track, 1 of 6 Selected) — May 2026
- Built an order-book matching engine and iterated from momentum, mean-reversion, and market-making strategies with a loss-limit kill switch.
- Outperformed the program's benchmark trader in the live reactive environment after analyzing Parquet market data using Pandas/NumPy.

Jane Street — **New York, NY**
Undergraduate Fellow (UNBOXED 2024; FOCUS 2025) — Jul 2024; May 2025
- Selected 1 of 37 (UNBOXED ’24) and 1 of 14 (FOCUS ’25); Wrote 150+ lines of SQL for data analysis and awarded a $2,000 scholarship.
- Achieved 15% return improvement in market simulation games through 30+ hours of coursework in statistics and market-making.

USC Information Sciences Institute (ISI) — HUMANS Lab — **Los Angeles, CA**
Undergraduate Research Intern — LLM-assisted AI for TikTok Eating-Disorder Dataset (EDTok) [arxiv] — August 2024 - May 2025
- Published EDTok, an ethics-compliant, multimodal TikTok dataset of 43,040 videos and 577,071 comments (2019–2024), raising dataset precision by 24% using a two-stage filter: weak-supervision rules + LLM zero-shot relevance classification with prompt engineering.
- Surfaced platform-scale insights on dataset (537M views, 79.9M likes, 962k shares across 10.9k users) via longitudinal analysis, BERTopic topic modeling, and multi-label emotion classification, linking themes to affect (e.g., recovery to optimism/joy; body-image to fear/sadness).

ACTIVITY / EXTRACURRICULAR - 3x Hackathon Winner

- **Awards:** 2026 Claude Hackathon Winner, 2025 Caltech HackTech Winner, 2024 HackHarvard Winner, 2024 FTC Robotics Regional Semifinalist
- **Selective Programs:** OpenAI Ambassador, D. E. Shaw Connect, Two Sigma New Seekers, Susquehanna (SIG) Discovery, Datadog's DataPUPS

PUBLICATIONS - [Google Scholar]

- C. Bickham, ***B. Ramirez-Gonzalez***, M.D. Chu, K. Lerman, E. Ferrara. *EDTok: A Dataset for Eating Disorder Content on TikTok*, **ICWSM 2025**

PROJECTS

Basis *(Award-Winning Agentic AI Cost-Segregation Platform @ **Fall 2025 USC LavaLab Winner** — github.com/bryanrg22/Basis_Info)*
Co-Founder & Lead Developer — September 2025 - Present
- Architected a human-in-the-loop agentic platform generating IRS cost-segregation studies in under 10 min on GCP (Cloud Run, Terraform IaC) and Temporal durable workflows, hardened with 3,500+ tests (chaos, contract, load), scaling throughput 15× (6.6 to 104 studies/hr).
- Built a vision pipeline pairing Claude/GPT VLMs with SAM 3 open-vocabulary segmentation, raising detection recall from 57% to 87% at 92% precision via task decomposition, per-class model routing, and eval-driven benchmarking of 26 vision architectures.
- Built an agentic hybrid RAG system (BM25 + pgvector HNSW, reciprocal rank fusion) over 7,466 chunks of IRS guidance with word-for-word citation verification, raising verifiable rule citations from 0% to 100% and cutting engineer review load from 91% to 50%.

iPhone Agent *(Computer-Use AI Agent for Physical iPhones via MCP @ **Claude Hackathon Winner** — github.com/bryanrg22/phone-control)*
Lead Developer — January 2026 - Present
- Built a computer-use agent harness that operates a physical iPhone through a perception–action loop (screenshots + accessibility-tree grounding to tool calls), completing real multi-step tasks and scoring 74% on hand-verified iOSWorld runs at under $0.01 per task.
- Shipped it as a live MCP server (OAuth 2.1/PKCE) callable from Claude, with an encrypted CoreDevice tunnel over Tailscale and pre-warming that cut task start-up 7.7× (37 s to 4.8 s), validated by 800+ unit checks and 36 fault-injection scenarios.
- Led cross-model evaluation of 7 frontier models across 640+ runs with a cross-family LLM-as-a-judge ($\kappa = 1.00$) and regression testing against baselines, human-auditing 243 runs to catch the judge’s 20% false passes and correct a published result.